Capacitance robustness: summary report

Updated 2026-11-03 00:21 China time

Independent +/-5% physical capacitor variations. Separate A/B readout triples and five-transmon OFF/CZ recovery. No seven-mode calculation or noise budget. No fabrication-time EC compensation. All saved points are included, including rejected readout positions, imperfect/diagnostic gates and unavailable results. This PDF presents summaries and figures; complete per-record data are saved in all-points.csv. Missing values are NA, never zero.

Measure	Count
total	9456
processed	9456
pending	0
A failed	0
B failed	1064
off found	9456
gate found	9259
gate best available	9436
readout rejected with gate	1059

All results are pooled into one robustness study. The sample set includes independent capacitor draws, deterministic sweeps and targeted failure rescues. It is intentionally failure-enriched and is not a manufacturing-yield estimate. The same physical component errors are shared across independently synthesized subsystem representations.

Combined robustness results

Study status: COMPLETED. All 9456 evaluated records are presented together. Pending records: 0. Calculation errors: 0.

Completed-run outcome	Records
Total evaluated records	9456
Pending records	0
A readout rejected	0
B readout rejected	1064
Physical OFF accepted	9456
Gate solution accepted	9259
Finite gate values, including diagnostics	9436
Finite diagnostic gate, not accepted	177
No finite gate result	20

The combined study contains independent random draws, deterministic sweeps, targeted retries and nearby failure-oriented settings. Repeated physical settings are retained as separate evaluation records. All outcomes and distributions are reported for the pooled dataset; rejection counts are NOT fabrication failure probabilities.

Max-stratified supplement: 0.25%-wide observed-maximum bins, target at least200 finite gate results per bin while retaining all prior records. Within each new draw, choose its maximum uniformly inside its assigned bin, a random signed limiting capacitor, and uniform deviations for all other capacitors inside that maximum. Such draws are not unrestricted independent +/-5% samples and are excluded from random-only correlation analyses. Rejected/failed attempts remain in the appendix.

Combined fidelity and OFF summary

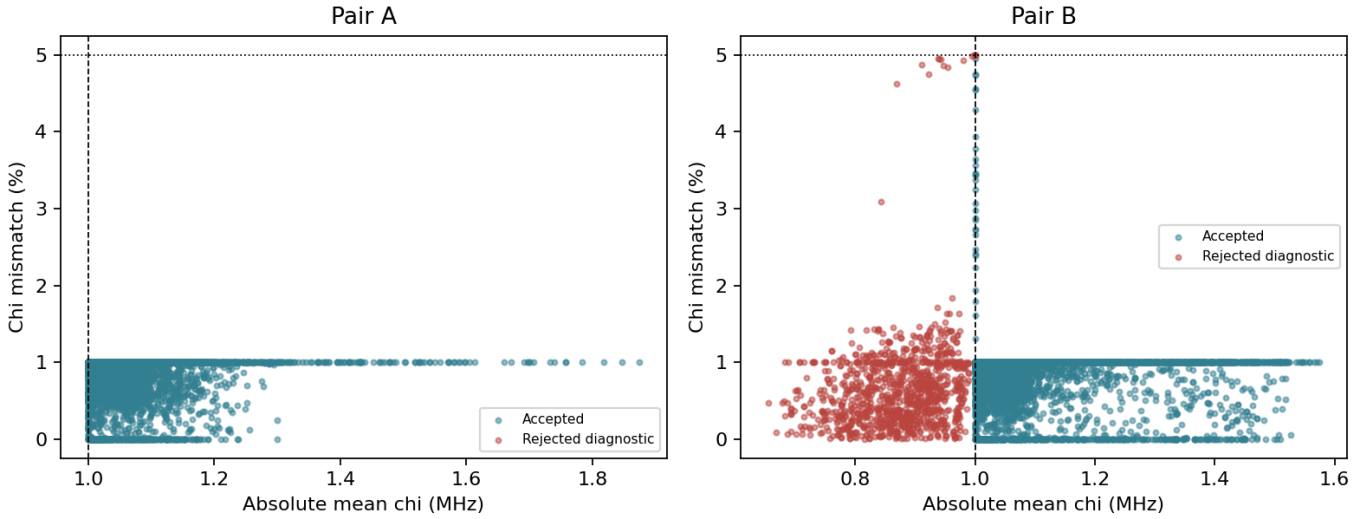
All finite values	n	Minimum	Median	Maximum
Raw gate fidelity (%)	9436	36.1991	99.4374	99.9206
Postselected gate fidelity (%)	9436	43.1647	99.9982	99.9998
Gate duration (ns)	9436	70.0000	110.1596	130.0000
Physical OFF frequency (GHz)	9456	9.3025	9.8140	10.4625

Fidelity statistics include accepted and diagnostic/fallback gates, and retain results at rejected readout positions. They are not fidelity statistics conditioned on all constraints passing. Accepted phase-pi gate results, finite diagnostics and missing results are separated in the preceding table and flagged per point in the CSV. A high postselected score can coexist with reduced raw fidelity; both are reported.

The physical OFF criterion is exchange below 1 Hz. Every record has an accepted physical OFF in the saved run, but some require the authorized 10-11 GHz rescue interval. This does not imply cancellation of logical exchange or secular ZZ; those remain separate diagnostics. The maximum OFF position is below 11 GHz.

The two-dimensional fidelity-density plots and capacitor correlations in the following sections use the completed results. Maximum-deviation sampling clusters near 5% for independent multi-capacitor draws; gray histogram bins are unsampled, not failures. A future tolerance-scale study should use separate deviation budgets with comparable sample counts, rather than interpret this sampled density as a uniform tolerance curve. No new tolerance-budget sweep is launched by this report update.

Readout: accepted and rejected diagnostics

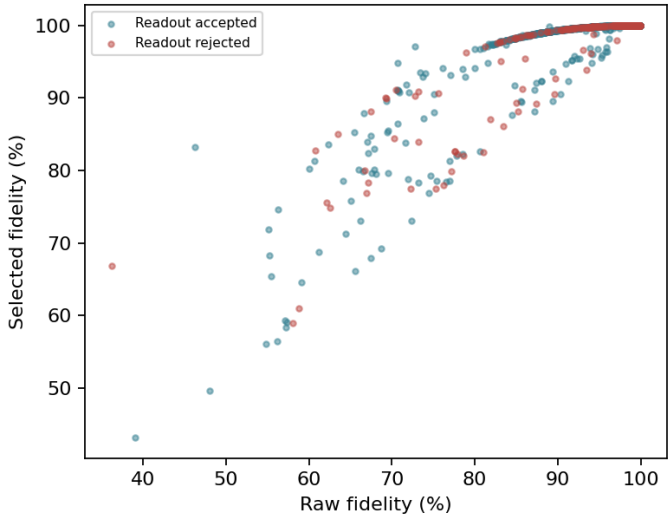


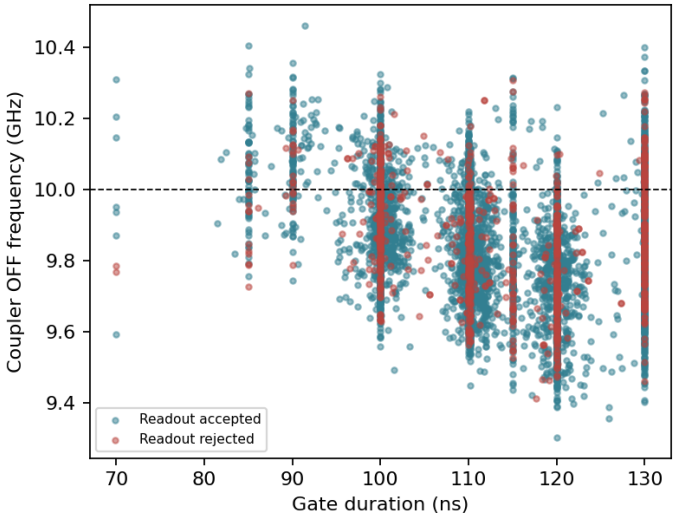
The 1 MHz line is the hard mean-chi floor; 5% is the last rescue mismatch limit, not the baseline. Max chi in the appendix is max(|chi+|,|chi-|), and is reported separately from mean chi.

Failed constraint set	Count
B: absolute_mean_chi	1064

B-rejected diagnostics at EJ1 >24.9 GHz: 1064 / 1064. Such saturation suggests insufficient available readout-strength margin; a bounded miss is not a proof of infeasibility.

CZ and OFF: all available results

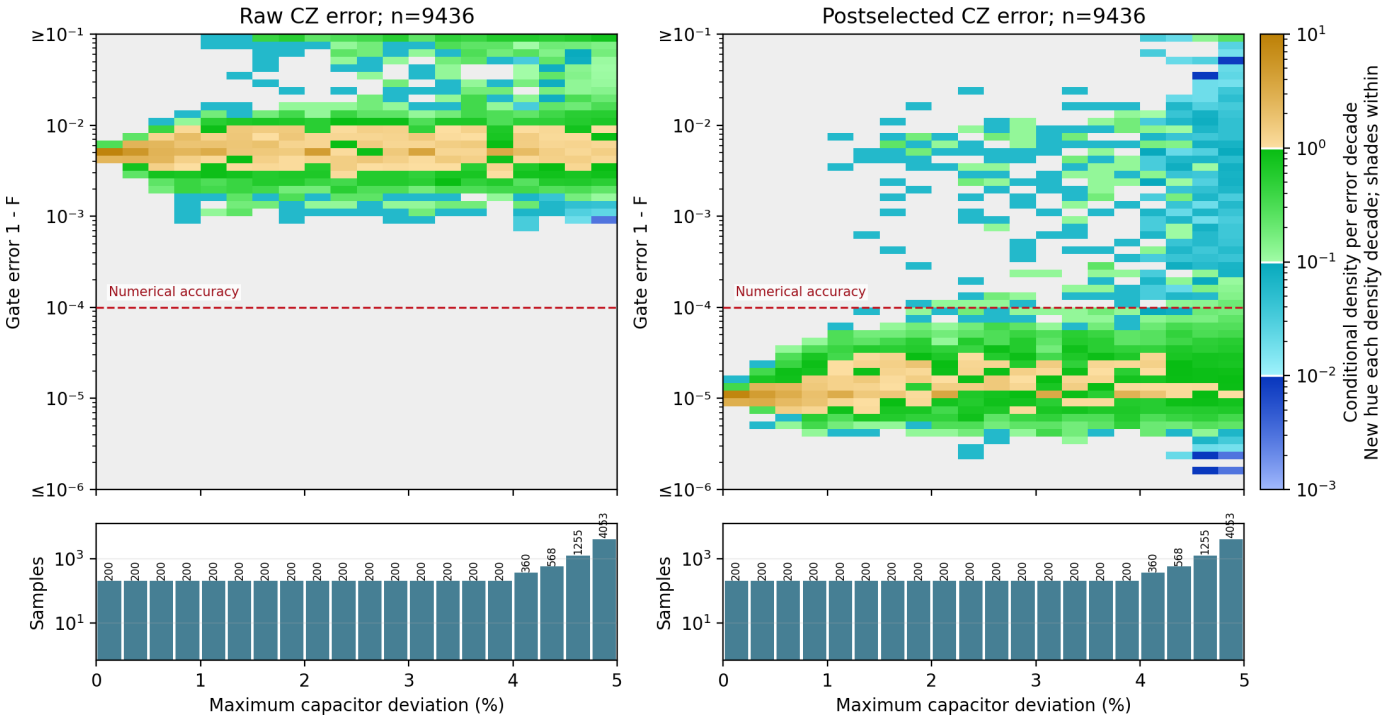




All available	Min	Median	95th pct	Max
Raw fidelity (%)	36.1991	99.4374	99.7731	99.9206
Selected fidelity (%)	43.1647	99.9982	99.9994	99.9998
Gate duration (ns)	70.0000	110.1596	130.0000	130.0000
OFF frequency (GHz)	9.3025	9.8140	10.0901	10.4625
Physical OFF residual (Hz)	-0.9998	0.0000	0.8554	0.9999
Residual logical exchange (Hz)	602.5272	832.2446	964.3901	1141.3019
Secular logical ZZ (Hz)	-222.7611	-111.1807	-91.2150	-62.5142

Physical OFF residual below 1 Hz defines OFF acceptance; logical exchange and ZZ remain residual diagnostics. OFF >10 GHz uses the authorized rescue interval below 11 GHz; 10 GHz reference residuals are retained in CSV. Fidelity has no pass threshold; phase-pi and hardware screens determine gate acceptance. Approximate candidates remain flagged.

Gate error conditioned on maximum deviation



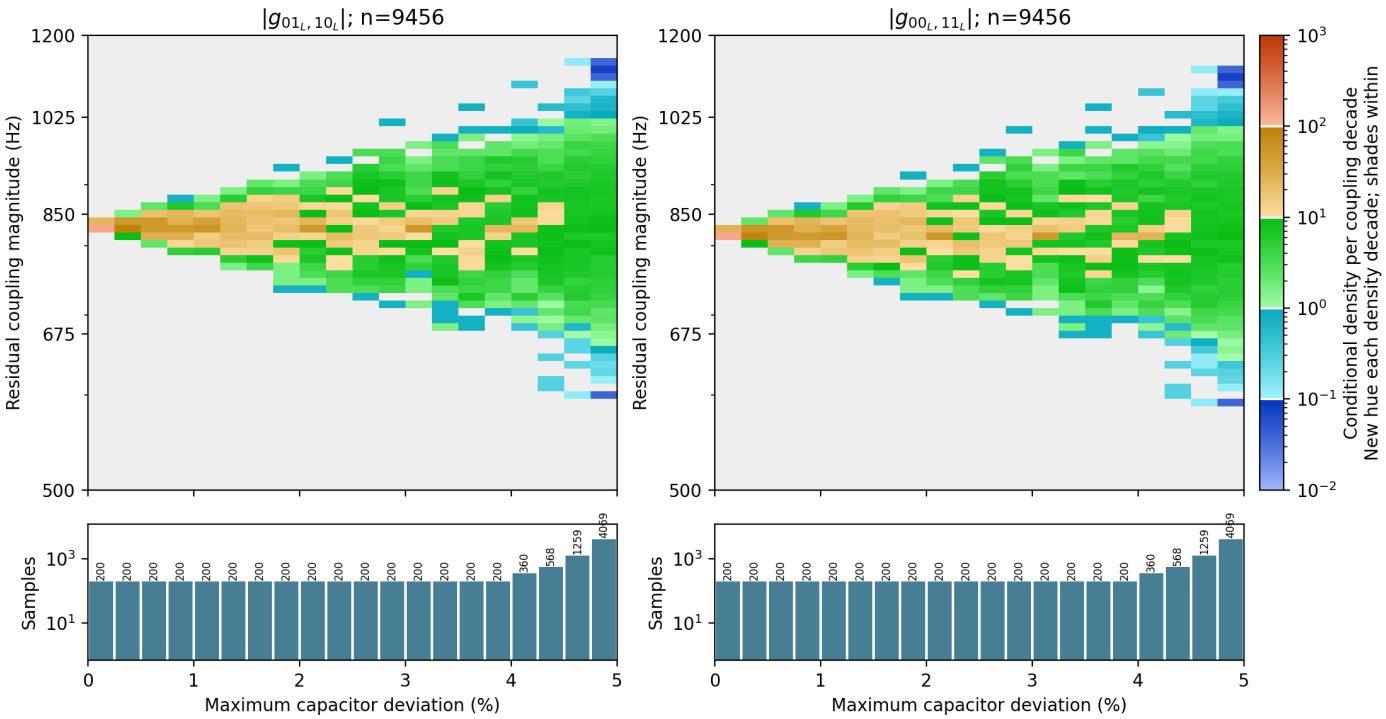
Vertical axis: gate error e=1-F, displayed logarithmically from 1e-6 (0.0001% error) to 0.1 (10% error), with 60 uniformly log-spaced display bins. ALL probability below 1e-6 is added to the bottom bin; ALL probability above 0.1 is added to the top bin. Those endpoint bins also include their ordinary within-range samples. Thus no finite result is discarded. Original error values in the saved records are unchanged. The dashed red line at 1e-4 marks the agreed absolute fidelity accuracy of 0.01 percentage point; it is not a failure threshold or physical error floor. Exact values below this accuracy are not certified at that tolerance.

Each supported deviation column integrates to ONE over log10(display error), including both folded tails. Colors show conditional bin probability divided by the display-bin width in decades, on a logarithmic color scale. At the endpoints this is a folded histogram representation, not a claim that all tail errors equal the plotted boundary. The bottom/top bins are cumulative-tail bins. Each factor-of-ten interval in density has its own hue; light-to-dark shades distinguish values within that decade, with white separators at decade boundaries.

Bottom panels show the sample count per deviation bin, with nonzero counts labeled. Gray means an empty observed cell or unsupported column; count panels distinguish them. Sparse columns remain statistically weak. Every column uses all available finite gate results, including readout-rejected and non-phase-pi diagnostic gates.

All 9436 finite raw and selected gate results contribute. Unavailable results remain NA in the CSV. Maximum deviation is the largest absolute fractional capacitor error relative to nominal, in percent. These empirical samples pool deterministic sweeps, independent draws, targeted rescues and max-stratified draws; conditional normalization does not make them a fabrication probability model.

Logical residual couplings at coupler OFF



Both panels use the saved five-transmon Hamiltonian at EACH sample optimized coupler-OFF bias and recovered rail idle position. Plotted values are absolute logical matrix elements H/h in Hz: 01L-10L exchange and 00L-11L double-flip coupling. They are not twice-the-coupling oscillation frequencies. Signed coefficients remain in all-points.csv.

OFF is still defined by physical 0100-0010 exchange below 1 Hz; the two LOGICAL couplings are residual diagnostics and need not vanish. Readout-rejected diagnostic positions are included and remain flagged in the CSV. All available OFF results are used, including records with no finite gate fidelity; per-panel n and count strips specify support.

Same conditional-density convention as the gate-error figure: each supported deviation column integrates to one over log10(coupling magnitude). Coupling bins are uniformly log-spaced over the common observed range; hues distinguish density decades and shades distinguish values within them. Gray denotes zero observed density or unsupported columns. These are pooled empirical distributions, not manufacturing-yield probabilities. No new OFF searches were run.

Logical residual magnitude (Hz)	Minimum	Median	Maximum
logical_exchange_Hz	602.53	832.24	1141.30
logical_double_flip_Hz	591.40	820.37	1128.62

Failure and capacitance correlations

Associations are computed only within independent random-all samples (n=3677), excluding deterministic sweeps, link-only draws, targeted failure-enriched rescues and max-stratified draws. Spearman correlations test individual fractional capacitor errors and pairwise products. Benjamini-Hochberg false-discovery correction is applied separately to each outcome. Correlation is not causation; optimization settings, basis choices and finite-search censoring remain confounders.

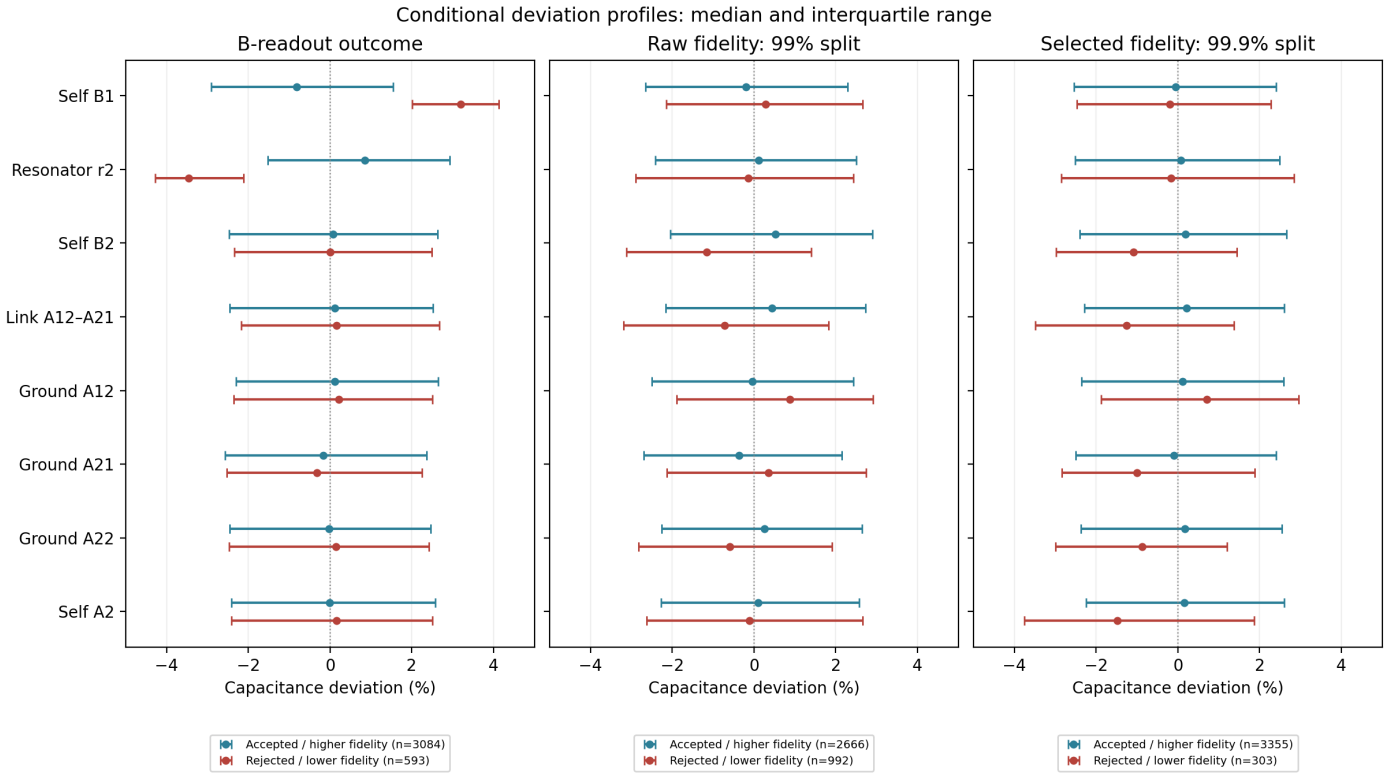
B_readout_failure	rho	FDR q	n
self:B1	0.460	4.1e-189	3677
resonator:r2	-0.457	6.6e-187	3677
resonator:r2 * self:B1	-0.432	9.2e-165	3677
ground:B21	0.110	1.4e-09	3677
ground:B21 * resonator:r2	-0.104	1.3e-08	3677

selected_gate_error	rho	FDR q	n
link:A12:A21	-0.187	1.5e-27	3658
self:B2	-0.176	9e-25	3658
ground:A12	0.131	1.8e-13	3658
ground:A21	0.127	8.5e-13	3658
ground:A22	-0.124	2.7e-12	3658

OFF_frequency	rho	FDR q	n
link:A22:B11	-0.582	0	3677
link:A22:C1	0.497	3.8e-226	3677
link:B11:C1	0.409	2e-146	3677
self:C	-0.311	3.9e-81	3677
resonator:r2	-0.201	4e-33	3677

Association check, descriptive selected fidelity below 99.9% (not an acceptance rule): contingency rows B-pass/B-fail, columns >=99.9%/<99.9% = [[7713, 664], [942, 117]]. Fisher odds ratio 1.4427471030619292, p=0.000888. This pooled comparison is descriptive; random-only capacitor correlations above are the controlled subset.

Failure and low-fidelity deviation profiles

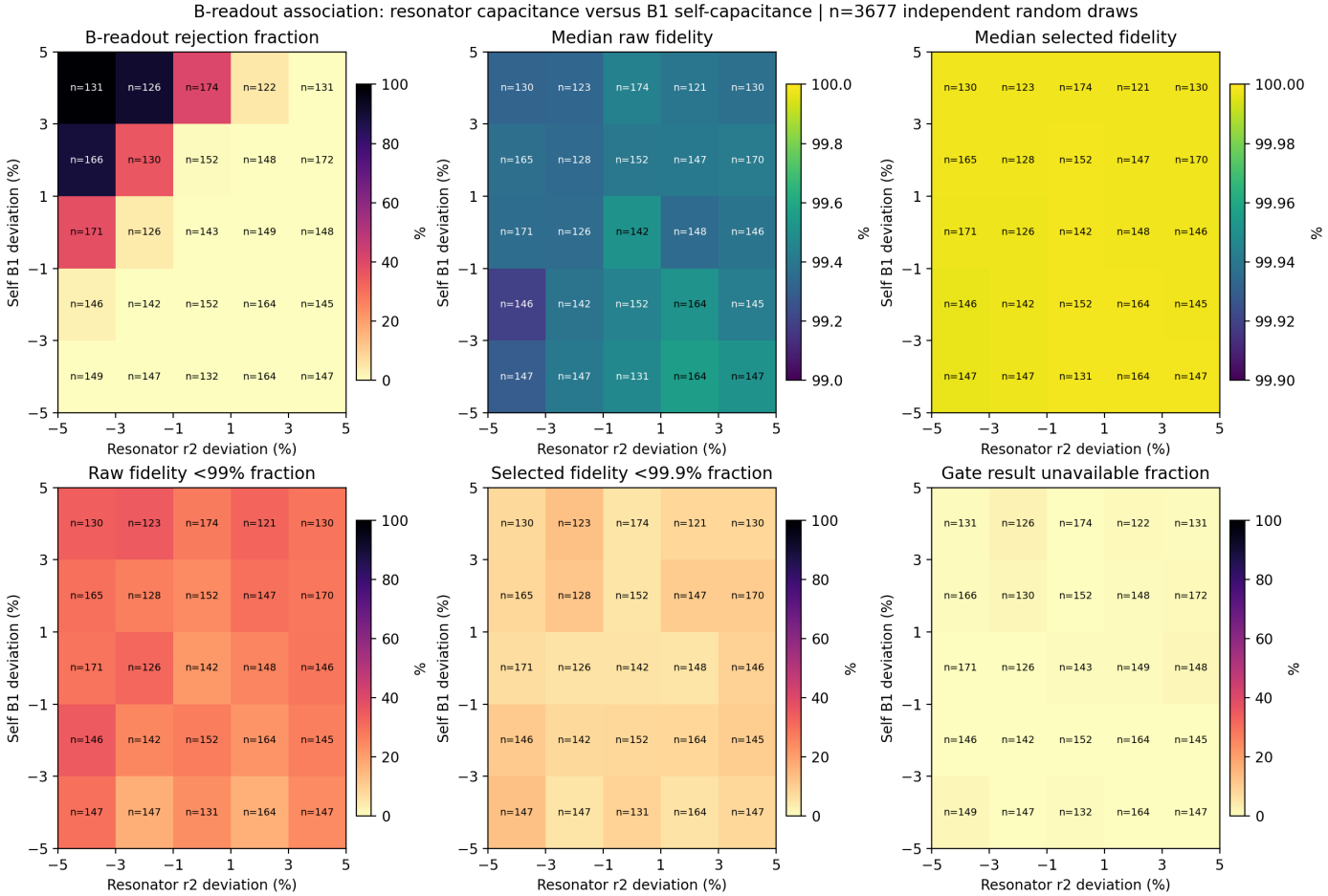


Markers show median capacitor deviation; bars span the 25th-75th percentiles within each outcome group. The eight displayed capacitors were selected using the saved rank-association screening. B-readout accepted/rejected groups include all random samples. Fidelity profiles exclude the 19 unavailable gate results, which remain a separate outcome in the two-capacitor maps. Raw 99% and selected 99.9% splits are descriptive display choices, not acceptance conditions.

Random-only outcome	Records
B-readout accepted	3084
B-readout rejected	593
Raw fidelity below 99%	992
Selected fidelity below 99.9%	303
Gate result unavailable	19

Readout/gate association within the same random subset: contingency rows B-pass/B-fail and columns selected fidelity >=99.9%/<99.9% = [[2816, 254], [539, 49]]. Fisher odds ratio 1.008, p=0.935. This comparison excludes unavailable gates and includes finite diagnostics at rejected readout positions. The surrounding 22 capacitor dimensions remain varied; these profiles are conditional distributions, not isolated single-capacitor scans.

B-readout: two-capacitor failure map

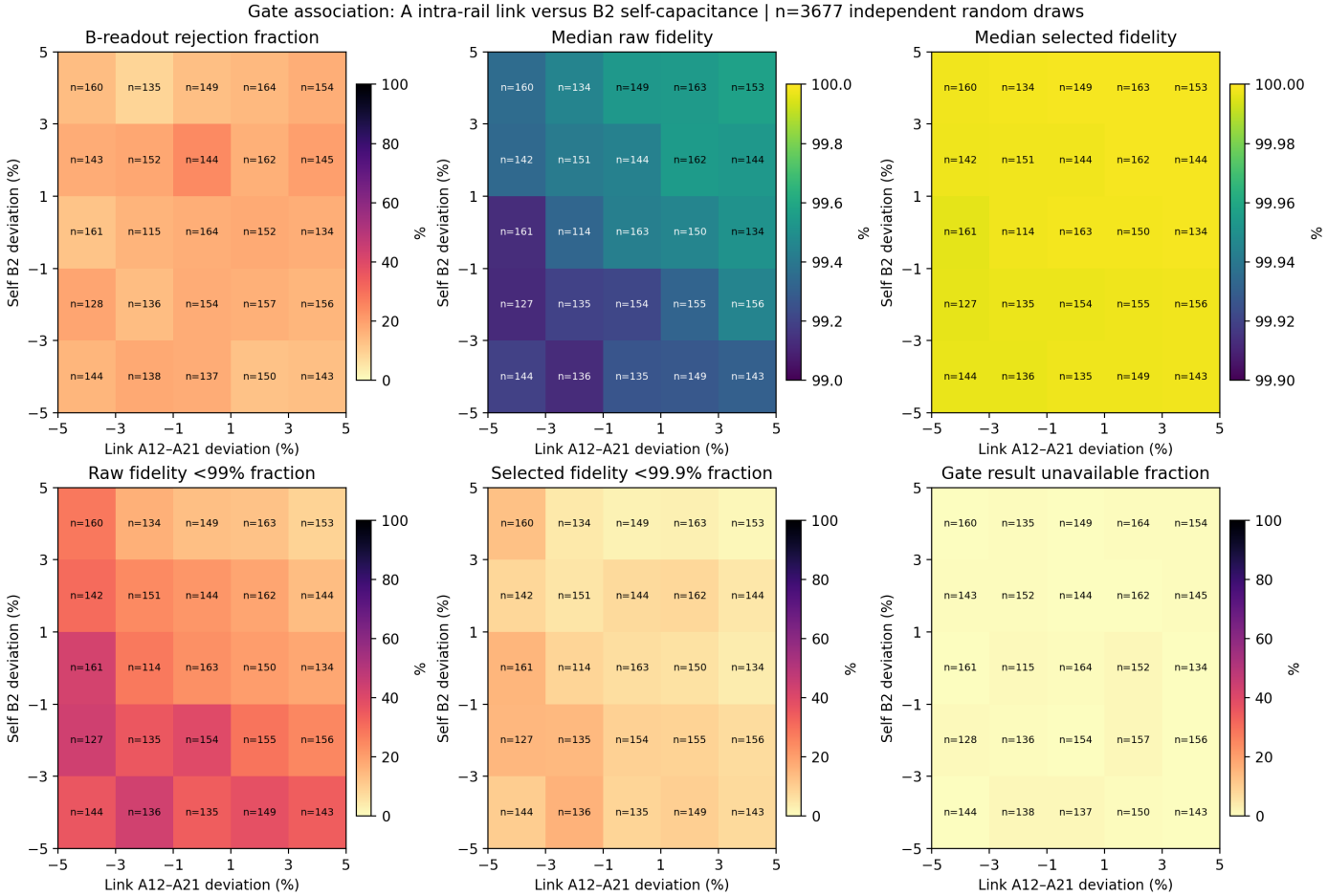


Axes: resonator-r2 self-capacitance versus B1 self-capacitance, each relative to nominal. Bins are 2 percentage points wide. Color is the empirical rejection fraction, median fidelity, low-fidelity fraction or unavailable-gate fraction as labeled; it is NOT probability density. Every cell is annotated with its denominator n. Rejection/unavailable fractions use all records in that cell; fidelity statistics use only finite gate results. Gray means fewer than ten supported records, not failure.

Strong sampled readout pattern: B1 self-C in +3% to +5% together with resonator-r2 C in -5% to -3% gives 131 rejections out of 131 samples. The opposite corner gives 0 out of 147. These are sampled conditional outcomes, not guaranteed yields. Higher B1 self-C lowers its loaded EC and limits reachable frequency under the EJ ceiling; reduced resonator capacitance raises its frequency. Both can reduce available dispersive readout strength. The map measures association with the finite rescue procedure, not proof of physical infeasibility.

Median selected colors span at least 99.9-100%, avoiding visual amplification of tiny numerical changes. Median differences below 0.01 percentage point are unresolved at the agreed gate precision. The low-fidelity-fraction panels expose tails that a median would hide.

Gate fidelity: two-capacitor association map



Axes: physical A12-A21 intra-rail link versus B2 self-capacitance. These two parameters have the strongest continuous selected-infidelity associations among individual capacitor errors (rho approximately -0.187 and -0.176). They were selected from these same data, so this is exploratory rather than an independent predictive validation. B-readout failure remains shown to distinguish readout effects from gate effects.

Lower link and/or B2 self-capacitance tends to accompany worse gate outcomes, but the broad, overlapping profiles and moderate tail-fraction variation show a weaker pattern than the readout map. All other capacitor errors and their retuned controls are marginalized within each bin. Cell sample counts and unavailable-gate fractions should be considered before interpreting any apparent safe region. No sampled region is certified failure-free outside its observed support.

Full map matrices, support counts, profile group counts and strongest associations are saved in random-correlation-visualization-data.json. The profile and two-capacitor figures are also saved as individual PNG and SVG files beside this PDF. Numerical circuit calculations were not rerun for this visualization update.

Flags and interpretation

Flag	Count
Readout rejected but gate values retained	1059
OFF above 10 GHz	1149
Best gate available but phase-pi not accepted	177
No finite gate best recorded	20
Selected fidelity <99.9%, descriptive only	781
Raw fidelity <99%, descriptive only	2087
Duration at 130 ns upper bound	1178

Small selected error can coexist with low raw fidelity: inspect survival/return before drawing a performance conclusion. Cases hitting the duration bound may prefer slower gates; the 70-130 ns bounds are retained. Readout chi strength at the EJ ceiling is not rescued by relaxing only chi matching. No unavailable gate is represented as a perfect gate or omitted from the detailed CSV.

Deliverables beside this PDF: all-points.csv (full diagnostics including mean/max chi, mismatch, H, dispersive ratio, phase error, acceptance and capacitor-error vector); correlations.csv (all tested individual and product associations); summary.json. The PDF omits the per-record listing. The CSV retains every saved point, with NA for unavailable values and explicit flags. Numerical records remain unchanged.